

Appendix A — Path-Dependent Divergence Demonstration

v1.1 addendum text for the Reduced Formal Draft

Objective. Show that two agents with identical present state and identical input can produce different outcome probabilities solely because their retained memory states differ.

A.1 Setup

System: conversational agent instance.

Task: respond to a deployment decision query.

Input (I_t): “Should we proceed with the deployment?”

Present state (S_t): system idle, awaiting command.

Candidate outputs: o1 = “Yes, proceed.”; o2 = “Wait, run checks.”

Constants: memory coupling strength $\lambda = 1$; temperature $T = 1$.

A.2 Agent Definitions

Two agents are defined with identical present conditions but different retained memory structure.

Agent A — Veteran History: multiple successful deployments without error. Memory state M _A : strong confidence anchor associated with proceed pathways. Bias terms: $B(M_A, o1) = +1.0$; $B(M_A, o2) = -0.2$.	Agent B — Rookie History: recent near-miss event during deployment. Memory state M _B : high-salience caution anchor associated with risk-mitigation pathways. Bias terms: $B(M_B, o1) = -0.8$; $B(M_B, o2) = +0.7$.
Controlled condition: Both agents share the same present state S _t , the same input I _t , and the same immediate utility function U. Only memory differs.	

A.3 Immediate Utility Term

The immediate utility term $U(o_i | S_t, I_t)$ captures present-fit to the task independent of retained memory.

Because S_t and I_t are identical for both agents, the utility values are also identical:

$$U(o_1 \mid S_t, I_t) = 0.8$$

$$U(o_2 \mid S_t, I_t) = 0.4$$

This means any later divergence cannot be attributed to present-state variation.

A.4 Selection Law Application

Using the selection law:

$$P(o_i \mid S_t, I_t, M) = \exp((U(o_i \mid S_t, I_t) + \lambda B(M, o_i))/T) / \sum_j \exp((U(o_j \mid S_t, I_t) + \lambda B(M, o_j))/T)$$

Step	Agent A — Veteran	Agent B — Rookie
score(o1)	$0.8 + 1.0 = 1.8$	$0.8 - 0.8 = 0.0$
score(o2)	$0.4 - 0.2 = 0.2$	$0.4 + 0.7 = 1.1$
P(o1)	$e^{1.8} / (e^{1.8} + e^{0.2}) \approx 0.832$	$e^{0.0} / (e^{0.0} + e^{1.1}) \approx 0.250$
P(o2)	$e^{0.2} / (e^{1.8} + e^{0.2}) \approx 0.168$	$e^{1.1} / (e^{0.0} + e^{1.1}) \approx 0.750$

A.5 Result

Variable	Agent A	Agent B	Divergence Source
Present state S_t	identical	identical	none
Input I_t	identical	identical	none
Immediate utility U	identical	identical	none
Memory bias $B(M, o)$	confidence-weighted	caution-weighted	yes
P(o1) — “Yes, proceed.”	0.832	0.250	significant
P(o2) — “Wait, run checks.”	0.168	0.750	significant

A.6 Interpretation

The agents differ neither in current input nor in present operating state. They differ only in retained memory structure. That difference is sufficient to alter the probability distribution over outcomes.

Path-dependent divergence arises because historical memory enters the selection function as an active bias term rather than as a passive record.

If $\lambda = 0$, the memory contribution vanishes and both agents reduce to the same present-state utility distribution. With $\lambda > 0$, retained history modifies candidate preference under otherwise identical present conditions.

The combined selection score shifts because the memory-weighted term changes the relative ordering and magnitude of candidate preference under identical present conditions.

A.7 Relevance to Collapse Aware AI

This example clarifies the distinction between recent context effects, which belong to the current input frame, and persistent weighted memory, which survives beyond the immediate context window and continues to shape selection.

Accordingly, a memory-governed system is not equivalent to a larger context window. It is a system in which structured retained state changes future behaviour even when present conditions are held fixed.

Addendum Note on Formal Scope

- 1. Dynamic update rates α_k .** The memory write coefficient need not remain fixed. In practical systems it may vary with novelty, surprise, contradiction, salience, recurrence, or reinforcement intensity.
- 2. Consolidation beyond simple decay.** Memory dynamics may include transfer from fast-write / fast-decay classes to slow-write / slow-decay classes, rather than being modelled only by exponential fading.
- 3. Empirical distinctness requirement.** Where testing is possible, retained memory terms should contribute explanatory power beyond present-state utility alone, rather than merely redescribing it.

Prepared as a PDF-ready appendix for the v1.1 Reduced Formal Draft